for Wheeler's team now to rewrite their computer programs so that, instead of simulating the explosion of a hydrogen bomb, they simulated the implosion of a massive star.

It would have been natural, that is, if the team still existed. However, the team was now disbanded; they had written their PMB-31 report and had dispersed to teach, do physics research, and become administrators at a variety of universities and government laboratories.

America's bomb design expertise was now concentrated at Los Alamos, and at a new government laboratory in Livermore, California. At Livermore in the late 1950s, Stirling Colgate became fascinated by the problem of stellar implosion. With encouragement from Edward Teller, and in collaboration with Richard White and later Michael May, Colgate set out to simulate such an implosion on a computer. The Colgate–White–May simulations kept some of Oppenheimer's idealizations: They insisted from the outset that the imploding star be spherical and not rotate. Without this restriction, their computations would have been enormously more difficult. However, their simulations took account of all the things that worried Wheeler—pressure, nuclear reactions, shock waves, heat, radiation, mass ejection—and did so by relying heavily on bomb design expertise and computer codes. To perfect the simulations required several years of effort, but by the early 1960s they were working well.

One day in the early 1960s, John Wheeler rushed into a relativity class at Princeton University that I, as a graduate student, was taking from him. He was slightly late, but beaming with pleasure. He had just returned from a visit to Livermore, where he had seen the results of the most recent Colgate, White, and May simulations. With excitement in his voice, he drew diagram after diagram on the blackboard, explaining what his Livermore friends had learned:

When the imploding star had a small mass, it triggered a supernova explosion and formed a neutron star in just the manner that Fritz Zwicky had speculated thirty years earlier. When the mass of the star was much larger than the 2-Suns maximum for a neutron star, the implosion—despite its pressure, nuclear reactions, shock waves; heat, and radiation—produced a black hole. And the black hole's birth was remarkably similar to the highly idealized one computed nearly twenty-five years earlier by Oppenheimer and Snyder. As seen from outside, the implosion slowed and became frozen at the critical circumference, but as seen by someone on the